

OPTIMAL STRATIFICATION UNDER TEST

Optimal Stratification of a Sampling Frame

Classical, Quantum, and Quantum-Inspired Approaches

A comparative study on the swissmunicipalities dataset — D-Wave, IBM Quantum, QCi Dirac-3, and Fixstars Amplify AE

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Optimal stratification of a sampling frame

Aggregate L atomic strata into K final strata so that the Bethel-Chromy allocation reaches a target precision (CV) with the smallest possible total sample size.



Combinatorial

The search space grows like K to the power L — exhaustive search is infeasible.



Not pairwise

The true objective is defined implicitly through the Bethel allocation, which couples every stratum in a cluster.



The shortcut

Replacing it with within-cluster homogeneity makes the problem tractable for QUBO solvers — at a price, quantified later in this deck.

L atomic strata $\rightarrow$ K final strata



↓ aggregate



minimise $n(\text{assignment})$ subject to CV targets

Objective: total Bethel sample size

$n = \sum \text{over final strata of the CV-optimal per-stratum allocation}$

The genetic algorithm in SamplingStrata

A population-based stochastic optimiser: chromosomes encode a cluster label per atomic stratum, and fitness is the true Bethel sample size — evaluated directly, not approximated.

Encoding

One cluster label per atomic stratum — a direct chromosome representation.

Fitness

The real Bethel-Chromy sample size, at every evaluation — no surrogate.

Search

Selection, crossover and mutation across generations; restarted to avoid poor local optima.

BEST-KNOWN RESULT

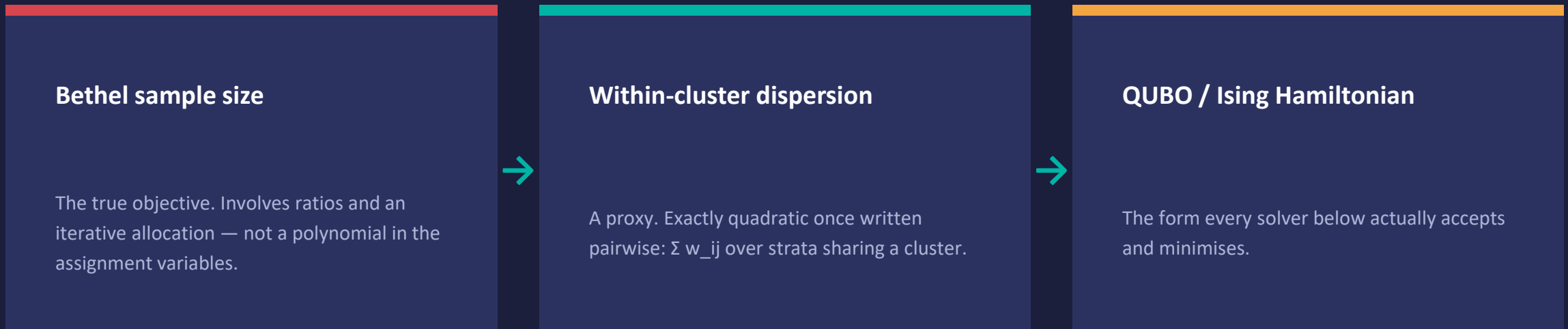
129

sample units at $K = 4$, on the twenty-stratum common frame used throughout this study

This is the benchmark every quantum and quantum-inspired result is measured against.

From clustering to a QUBO Hamiltonian

Every quantum and quantum-inspired solver in this study minimises the same surrogate: within-cluster dispersion, the one clustering objective with an exact quadratic, pairwise rewriting.



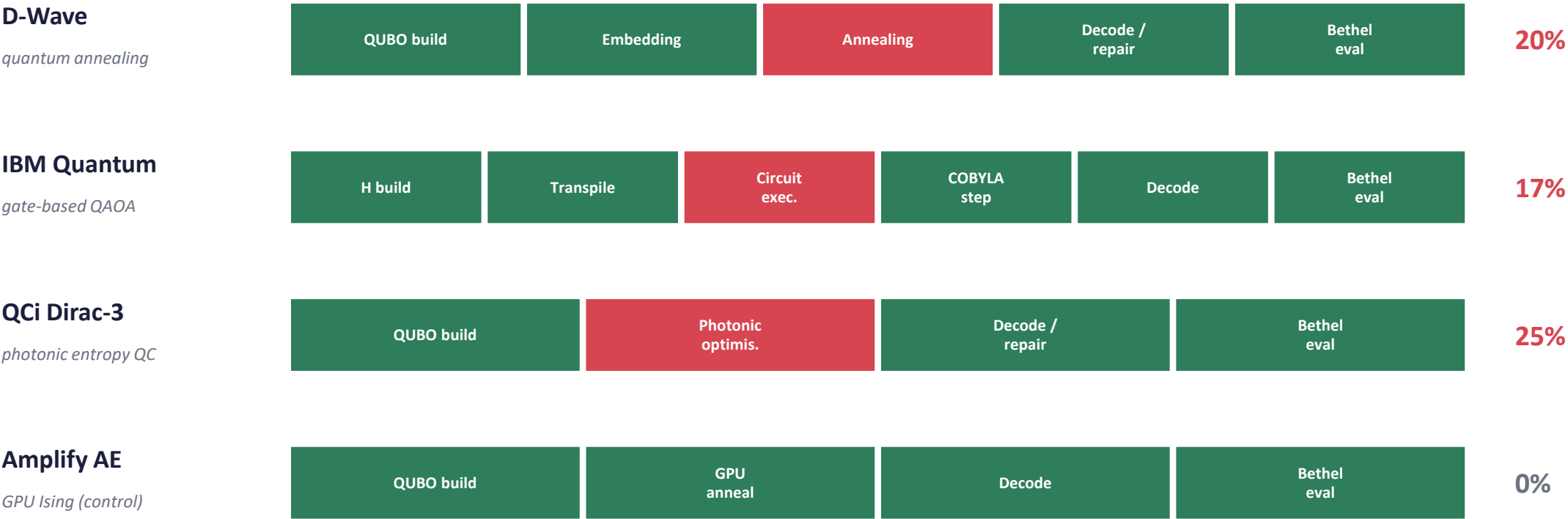
penalty = $3 \times \max(w)$ enforces the one-hot constraint (each stratum in exactly one cluster) across every one-hot formulation used here.

Same surrogate, four very different machines

D-Wave	IBM Quantum	QCi Dirac-3	Amplify AE
Quantum annealing	Gate-based QAOA	Photonic entropy QC	GPU Ising engine (control)
ENCODING	ENCODING	ENCODING	ENCODING
One-hot QUBO. Dense connectivity needs minor-embedding onto a sparse Pegasus lattice — chains of physical qubits per logical variable.	Binary-label encoding (not one-hot): $b = \lceil \log_2 K \rceil$ qubits/stratum. Dense interactions force costly SWAP routing on the heavy-hex lattice.	One-hot QUBO. All-to-all connectivity: no embedding, no SWAP routing, no variational loop.	Identical one-hot QUBO to D-Wave/Dirac-3. Free, no quota limits, no embedding needed.
QUANTUM SHARE	QUANTUM SHARE	QUANTUM SHARE	QUANTUM SHARE
Annealing itself is quantum; embedding, penalty, decode & evaluation are classical.	State prep & energy estimate are quantum; the COBYLA loop, transpilation & decoding are classical.	Optimisation is performed on photonic hardware per QCi's own description; construction & evaluation remain classical.	Zero. A purely classical control that isolates the cost of the formulation from the cost of quantum hardware.

How much of each pipeline is actually quantum?

Each bar is one solver's workflow, split into its computational stages, sized equally within that solver's own pipeline — this is a stage-by-stage schematic, not a measured runtime split.



Genuinely quantum step

Classical step

Note: Dirac-3 shows the largest nominal quantum share — yet the worst result (286). Quantum share does not predict solution quality.

Why the common frame stops at twenty strata

Not a statistical choice — an operational one. Twenty atomic strata is the largest problem every platform's free tier can accept, so the comparison is genuinely like-for-like.



IBM

Circuit depth, not qubits

40 qubits fit easily, but SWAP routing on the sparse heavy-hex lattice inflates the real $K=4$ circuit to ~5,354 two-qubit gates — already noise-dominated.



QCi Dirac-3

100-variable free-tier cap

The one-hot QUBO needs $L \times K$ variables. At $K=4$, the cap of 100 variables is reached at $L=25$ strata.



D-Wave

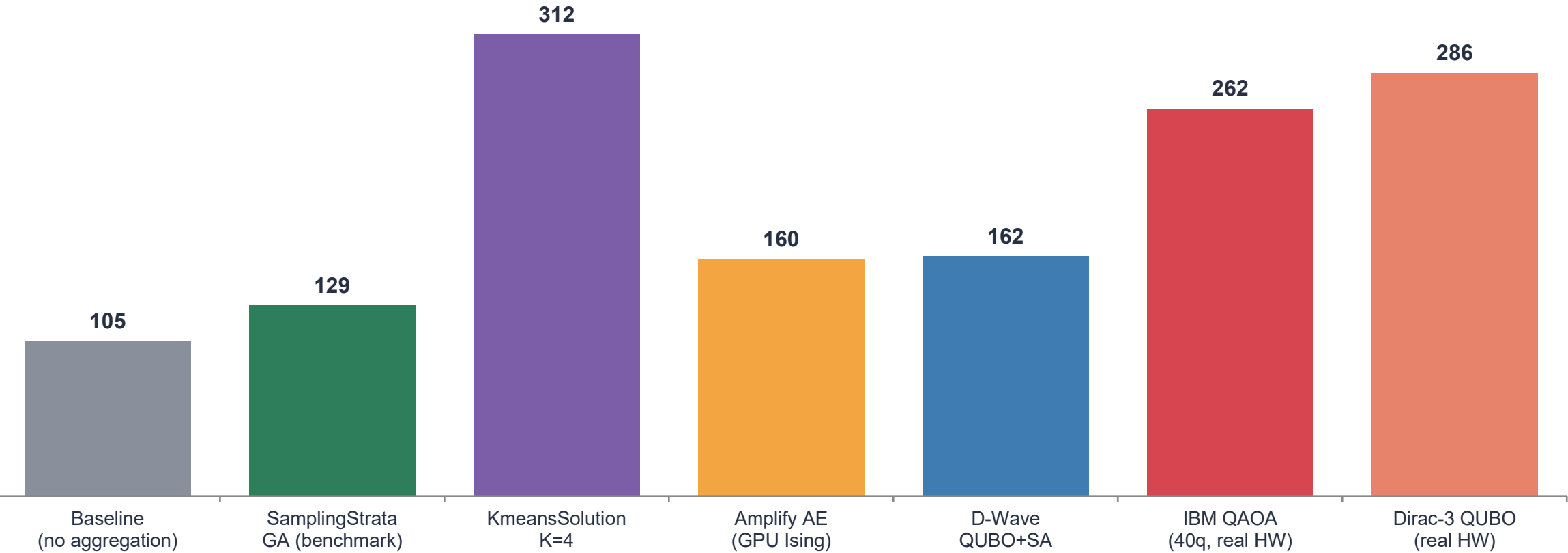
Minor-embedding capacity

The dense QUBO needs long qubit chains on Pegasus; practical embedding tops out around 120 dense variables — roughly 30 strata at $K=4$.

Twenty strata: fine enough to be a genuine optimisation problem, coarse enough to run on every platform without approximation.

Sample size by method — lower is better

Twenty atomic strata · $K = 4$ · CV target 5% · identical Bethel-Chromy evaluation for every method



● Classical benchmark (129) ● Best-known QUBO value — Amplify AE (160)

Two quantum machines, two different failures



IBM — noise, not optimisation



40-qubit circuit reaches depth ~4,900 after SWAP routing



Cost energy sat at zero throughout the (2-iteration) variational loop



No bitstring repeated across the sample taken



Best read: an effectively random partition, from a run too short and too deep



Dirac-3 — a real optimum, in the wrong place



Final energy clearly negative — genuine optimisation, confirmed



All-to-all photonic connectivity: no embedding, no routing overhead



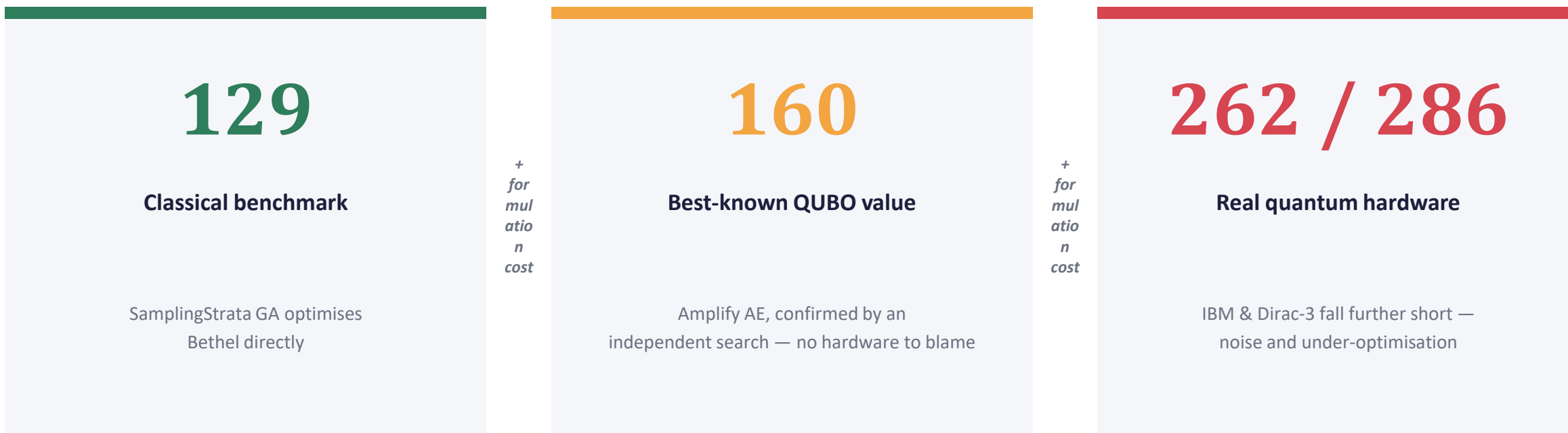
Yet it settles far from the best-known QUBO value (286 vs. 160)



It optimised the right surrogate — just not nearly well enough

Decomposing the gap: formulation vs. hardware

A free classical GPU (Amplify AE) isolates the best-known value the QUBO surrogate can reach — no noise, no embedding, no routing to blame.



The 24% gap from 129 to 160 is the price of the surrogate itself — no solver in this study, quantum or classical, closed it.

Why minimise a proxy, and not Bethel itself?

A QUBO must be a polynomial in the assignment variables. The Bethel sample size cannot be written that way.

The obstacle

- Cluster means are ratios: population appears in the denominator
- The denominator itself depends on the assignment — not a fixed constant
- Bethel-Chromy is an iterative fixed-point procedure with min/max truncation, not a closed formula
- None of this is expressible as a low-degree polynomial

Why homogeneity works

- Within-cluster dispersion has an exact pairwise identity in the assignment variables
- It is quadratic — exactly the form a QUBO needs
- The trade-off: it is an approximation of Bethel, not an equivalent of it
- This approximation, not any solver's weakness, sets the ~160 floor seen in this study

The wider landscape: what else was considered

Quantinuum / IonQ	<i>Trapped ion, all-to-all</i>	NOT USED — worth running	No SWAP overhead; the one experiment that could isolate connectivity from the surrogate problem
Fujitsu DA / Toshiba SQBM+	<i>Dense-QUBO hardware</i>	NOT USED — redundant	Would likely match, not beat, Amplify AE's best-known result
Google Willow	<i>Superconducting</i>	NOT AVAILABLE	Access closed at time of writing — selective research proposals only
Rigetti / IQM	<i>Superconducting, sparse</i>	NOT USED — unsuitable	Same sparse-lattice SWAP problem that limited IBM
QuEra / Pasqal	<i>Neutral atom, analog</i>	NOT USED — unsuitable	Would need a lossy remapping to a maximum-independent-set problem

CONCLUSIONS

Classical still wins — for two distinct reasons

129

Classical benchmark

160

Best-known QUBO value
(a free GPU, not a quantum device)

262 / 286

Real quantum hardware
(IBM / Dirac-3)

- › The surrogate objective, not any particular machine, sets the ~24% floor above the classical optimum.
- › Both quantum devices fall short even of that floor — for different reasons: IBM by noise, Dirac-3 by under-optimisation.
- › Two structural obstacles remain before quantum optimisation can compete: a Bethel-compatible objective, and hardware that executes dense interactions without noise.
- › Practical recommendation today: the SamplingStrata genetic algorithm remains the method of choice.